

Chengcen(Ryland) Fu

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CAREER OBJECTIVE

Full-time (After Graduation) — Simulation Engineer, also open to Robotics Software Engineer roles.

EDUCATION

University of Bristol <i>MSc, Robotics</i>	2026.09 – 2027.09
Beijing Film Academy <i>MFA, Film · Average grade: 89/100 · Third-Class Scholarship, 2023–2024</i>	2022.09 – 2026.07
Sichuan Fine Arts Institute <i>BFA, Animation · Average grade: 83.5/100</i>	2016.09 – 2021.06

WORK EXPERIENCE

National Centre of Applied Mathematics (Chongqing) <i>Simulation Development Intern — Intelligent Transportation Simulation</i> 1. Built a synthetic-data generation pipeline combining SUMO traffic simulation with Blender rendering: SUMO-generated traffic flows were imported into Blender with pre-built 3D vehicle and scene assets, then rendered into RGBA image sequences via a virtual camera to produce realistic training data. 2. Optimised the rendering pipeline from ray tracing to rasterization, cutting per-frame render time from 20s to 1s — shifting the workflow from offline batch data collection to real-time, on-the-fly training data generation. 3. Integrated imgaug-based augmentation into the rendering output to add weather (fog, snow) and camera-artefact variation, improving the diversity and robustness of the training data. 4. Researched reinforcement-learning approaches for adaptive algorithm selection within traffic-simulation scenarios.	2025.06 – 2026.01
INF Technology (Shanghai) Co., Ltd. <i>3D Art Intern</i> 1. Built real-time character–object interactions in Unreal Engine 5, where app events trigger animations, and created supporting VFX with Niagara. 2. Cleaned and retargeted motion-capture data in Maya; designed and produced character animation.	2024.05 – 2024.09

PROJECTS

RPG Game Demo (Unreal Engine 5) Implemented character movement, an animation state machine and interactive UI (clickable inventory and portrait bar) in C++ on top of Unreal's Actor and Component classes, using Live Coding for fast iteration.	2025.02 – 2025.04
Distributed System based on the Raft Protocol Implemented core Raft mechanisms — leader election and log replication — with RocksDB integrated for data persistence.	2024.06 – 2024.09
Memory Pool Implemented an memory pool in C++ to address memory fragmentation and reduce performance overhead.	2024.09 – 2024.12

SKILLS

Simulation & Engines: SUMO, Blender, Unreal Engine 4/5, Unity, imgaug
Programming & Systems: C/C++, Python, Linux (Ubuntu 22), GCC/GDB, CMake, VS Code, Visual Studio
Distributed Systems & Middleware: Raft, RocksDB, Redis, TCMalloc
3D & Visual: Maya, ZBrush, 3ds Max, Substance Painter, Marvelous Designer, Marmoset Toolbag, Adobe After Effects / Premiere, Photoshop